Biochemistry Cumulative Examination

March 5, 1994

Molecular biology is in the news again. The 1993 Nobel Prizes for Chemistry and Medicine were both given to workers either discovered something new about DNA or figured out new and better ways to manipulate DNA. Answer the following questions about their discoveries.

1. 50 points. Philip Sharp, now at MIT, and Richard Roberts, now at New England Biolabs, were awarded the prize in medicine. News reports usually said something to the effect that they independently discovered that there is extra DNA in genes. (News reporters usually are not trained in science.) Describe what their discover was in some detail. Give a general description and also answer the following specific questions: What organisms is this kind of "extra DNA" found in? Where in their DNA is it found? Describe the possible biological function or consequences of this "extra DNA" and include the names a biochemist would use for it.

2. 25 points. Micheal Smith also shared the prize in chemistry. News reports inform us he learned how to make specific changes in DNA or something vaguely along those lines. What did he discover? How does one use this technique (as much practical detail as possible, please) and what are its implications?

3. 25 points. Kary Millus shared the prize in chemistry. News reports invariably described how he went surfing after learning of the award. They were a little sketchy on what he did to get the prize, usually saying something to the effect that he learned how to copy bits of DNA. They always mentioned the movie *Jurassic Park* as well. (Surfing dinosaurs?) What did he discover? How does one use this technique (as much practical detail as possible, please) and what are its implications?

4. For NO bonus optional extra credit points: Perhaps the way to get a Nobel Prize is to get a movie made about your work. "Steven Speilberg presents *Electron Transfer*." Smuggling electrons across the border no doubt. Whose research project from this department should be made into a movie? What should it be entitled? And, most importantly, which actor(s) should play the role of the professor(s)? For example, Bruce Lee as Julian Pranata in *Komputational Kung-fu*. Interesting (not insulting) answers will be posted.